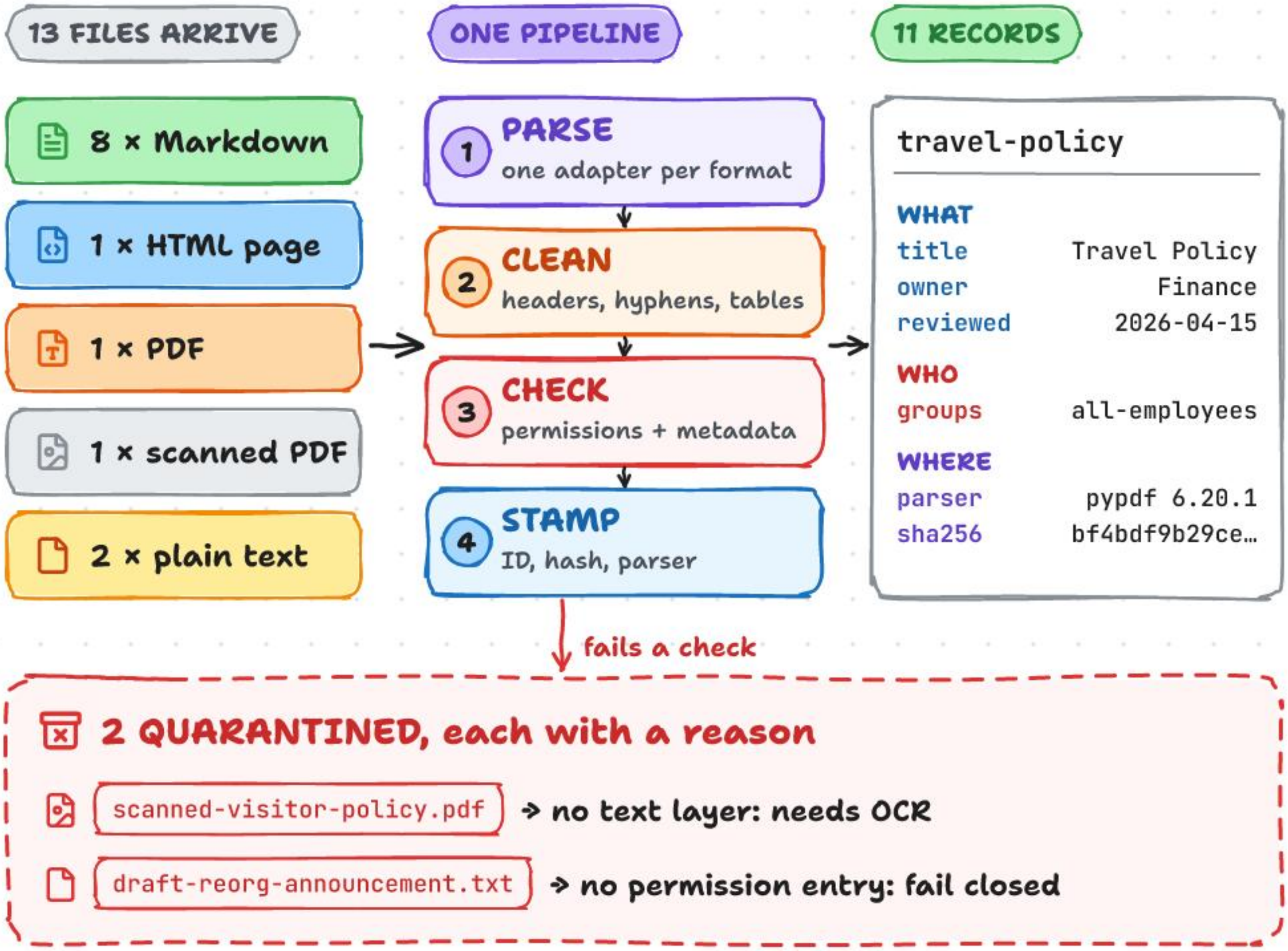
Messy files in, trusted records out

Day 02: ingestion, parsing and metadata, before any retrieval.



13 FILES ARRIVE

- 8 × Markdown
- 1 × HTML page
- 1 × PDF
- 1 × scanned PDF
- 2 × plain text

ONE PIPELINE

1. **PARSE** one adapter per format
2. **CLEAN** headers, hyphens, tables
3. **CHECK** permissions + metadata
4. **STAMP** ID, hash, parser

11 RECORDS

`travel-policy`

WHAT	
title	Travel Policy
owner	Finance
reviewed	2026-04-15
WHO	
groups	all-employees
WHERE	
parser	pypdf 6.20.1
sha256	bf4bdf9b29ce…

fails a check

2 QUARANTINED, each with a reason

`scanned-visitor-policy.pdf` → no text layer: needs OCR

`draft-reorg-announcement.txt` → no permission entry: fail closed

Retrieval can only find what ingestion kept.
So check what goes in, and say what was left out.

Real run · Day 02 notebook · free, no API key

Where each field comes from

Who can change a field decides how far you can trust it.

THE DOCUMENT ITSELF

front matter · <meta> tags · PDF properties

`title` `owner` `last_reviewed`

⚠ **Any author can edit it. Validate it.**

e.g. intranet page: review overdue (470 days)

THE SYSTEM OF RECORD

the permission export, not the file

`allowed_groups` **authoritative · missing = fail closed**

A dropped-in draft says:

"Audience: all staff (please share widely)"

The export says:

no entry →

visible to nobody

THE PIPELINE

added by your code on every run

`doc_id` `source_path` `parser` `content_hash`

Traces any answer to one file, one parser version and one exact version of the text.

PDFs store ink, not tables

One page of travel-policy.pdf, extracted two ways by pypdf 6.20.1.

WHAT YOU SEE

City	Hotel per night	Meals per day
London	USD 250	USD 75
New York	USD 300	USD 75
Singapore	USD 220	USD 60
Other cities	USD 180	USD 50

DEFAULT ("plain") MODE

follows the drawing order

```
City  London  New York  Singapore
Other cities  Hotel per night
USD 250  USD 300  USD 220
USD 180  Meals per day  USD 75
USD 75  USD 60  USD 50
```

One stream: all cities, then all hotel limits, then all meal limits.

LAYOUT MODE + CLEANING

places text by position

```
City | Hotel per night | Meals per day
London | USD 250 | USD 75
New York | USD 300 | USD 75
Singapore | USD 220 | USD 60
Other cities | USD 180 | USD 50
```

✓ **Rows survive: Singapore → USD 220**

claim reim- / bursement

re-joined → phrase found

header + footer on each page

4 lines → 0

Same document, new version

Two IDs, two questions. Compare them on every run.

WHICH document?

`doc_id` stays when the text changes

`remote-work-policy`

WHICH version?

`content_hash` changes with 1 character

USD 500 `9d0e7b74370d6210…`

USD 600 `e6e17cd1b3d6f8db…`

Real run: 5 changes to a copy of the data, then re-ingest

ADDED	`annual-leave-policy`	→ chunk + index
TEXT CHANGED	`remote-work-policy`	→ re-chunk + re-index
PERMISSIONS CHANGED	`password-and-mfa-standard +contractors`	→ update metadata only
REMOVED	`vpn-access-guide`	→ DELETE its chunks
UNCHANGED	`8 docs (one re-saved)`	→ skip

One file was re-saved with Windows line endings: new bytes, same text, so it is skipped.

Next → Day 03: chunking and document structure